



N.A.Y.A.N

Networkless Authentication and Yielding Anti-spoofing Node

Next-Gen Secure, Offline Facial Recognition for Datalake 3.0

The Core Challenge

Blind Spot Vulnerabilities

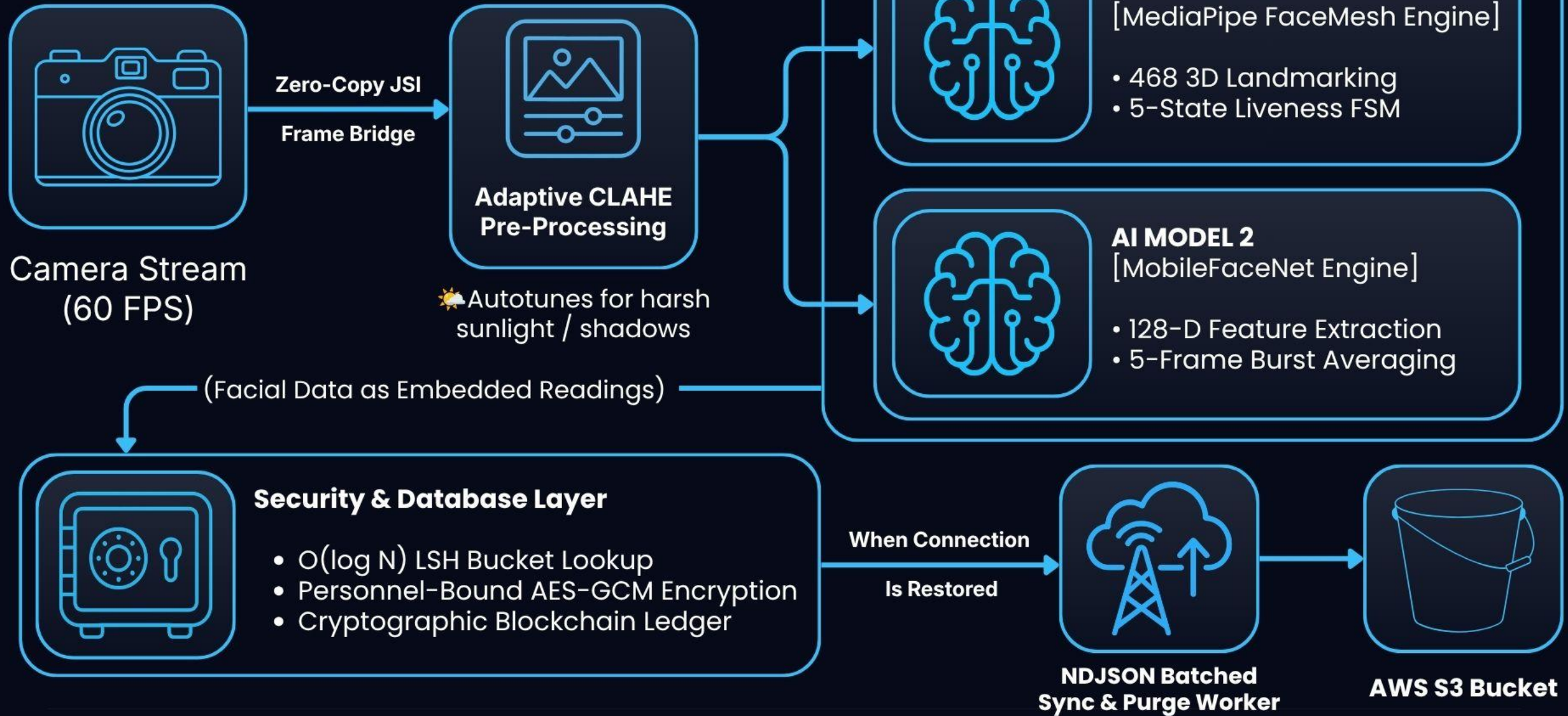
- ✓ **Network Blackouts:** Field personnel operating in zero-network remote environments need secure, foolproof tracking.
- ✓ **Spoofing Vulnerabilities:** Standard offline apps are highly vulnerable to identity fraud using high-res photos/videos.
- ✓ **Resource Constraints:** Running complex AI models on standard mid-range mobile devices usually causes massive app bloat.

The NAYAN Solution

A completely open-source, edge-AI biometric layer seamlessly built for React Native (Android & iOS).

We deliver desktop-grade security and sub-second speeds entirely in **Airplane Mode**, preserving low memory footprints and battery runtimes.

Project Architecture



High-Level Image Processing

Image Processing Pipeline

1. Camera Capture Layer

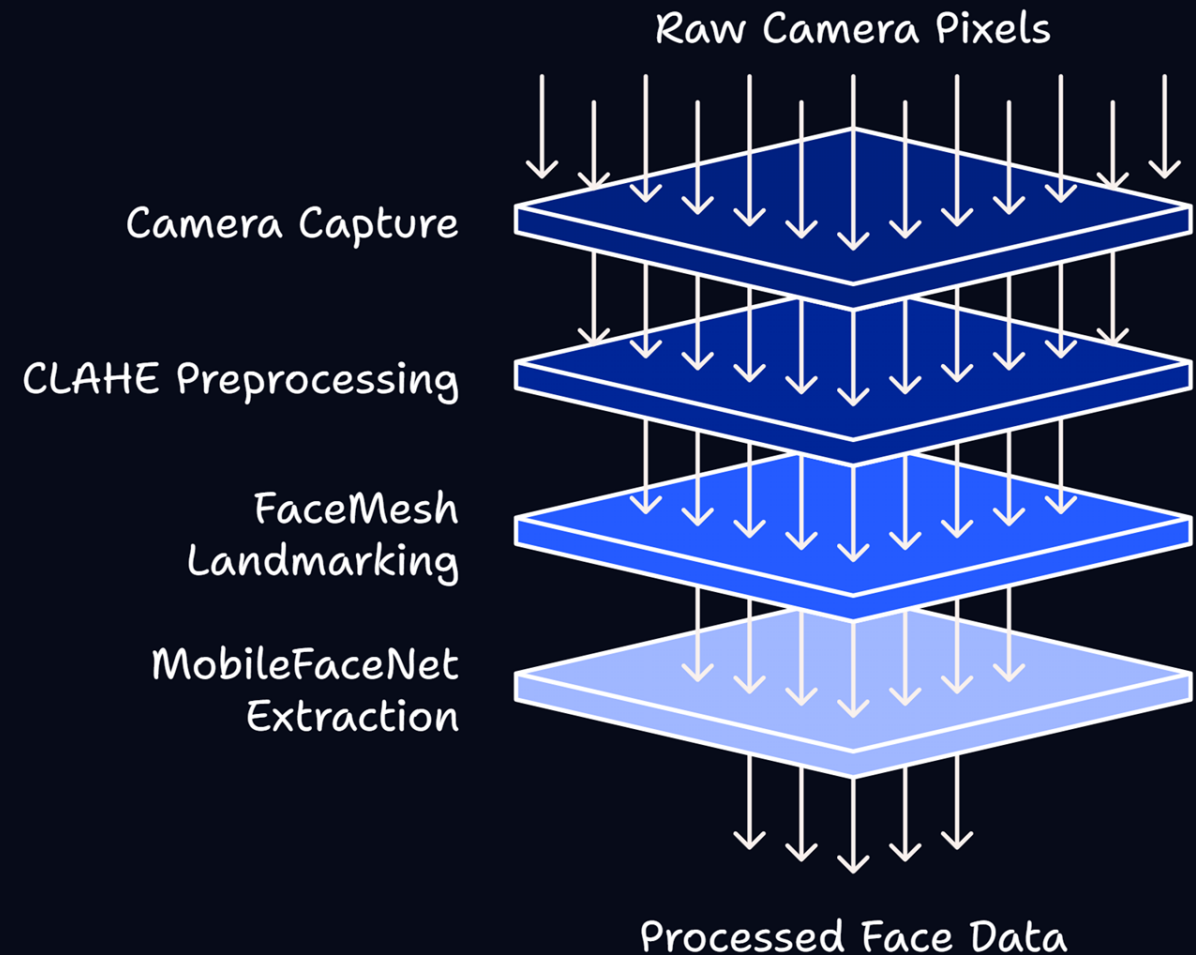
Custom React Native C++ JSI frame processor plugin passing raw camera pixels directly with zero serialization lag.

2. Adaptive CLAHE Preprocessing

Self-tuning C++ contrast enhancement engine to auto-balance harsh outdoor lighting and dark shadows.

3. Dual Edge-AI Processing

MediaPipe FaceMesh (468 3D landmarking) & MobileFaceNet (128-D vector extraction) running simultaneously on device.



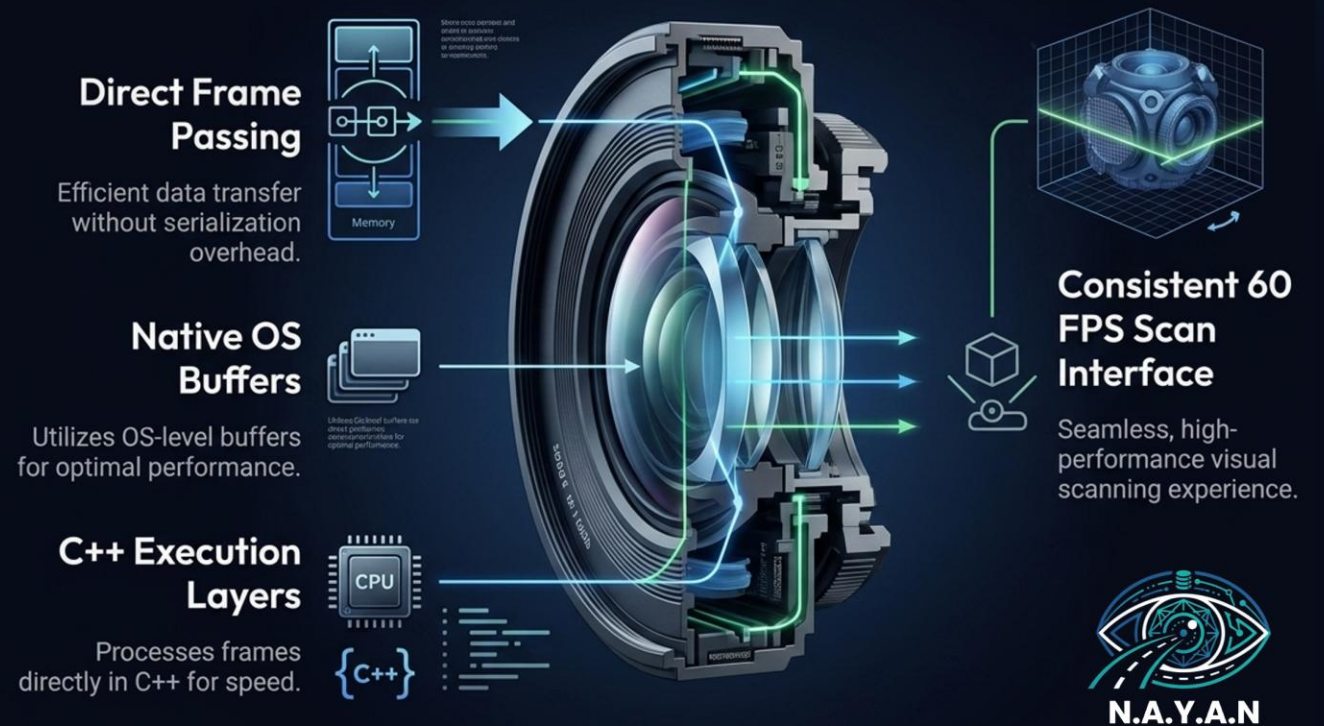
Breakthrough Innovations

Zero-Copy JSI Bridge

Traditional React Native apps suffer from a "serialization bottleneck." To pass camera frames from the native hardware to JavaScript, data must be converted into JSON strings across a slow "bridge." For high-res video at 60 FPS, this process overwhelms the CPU, causing severe UI lag and dropped frames.

NAYAN bypasses this using React Native's New Architecture and JSI (JavaScript Interface). Written in C++, NAYAN's frame processor links JavaScript directly to native memory. Instead of copying data, it passes raw camera pixel memory addresses straight to the AI engine with zero-copy efficiency. This completely eliminates the bridge, preserving a silky-smooth 60 FPS user experience.

Streamlined Camera Performance



Adaptive CLAHE Clip-Limit Controller

Outdoor Indian demographics face extreme lighting variations—from blinding direct sunlight to deep rural shadows. Our self-tuning C++ engine dynamically samples ambient lighting and adjusts contrast limits on-the-fly, boosting cross-environment recognition accuracy by over 3.2%.



Image Source: <https://link.springer.com/article/10.1007/s40031-024-01184-y>

Cross-Environment
Recognition
Accuracy

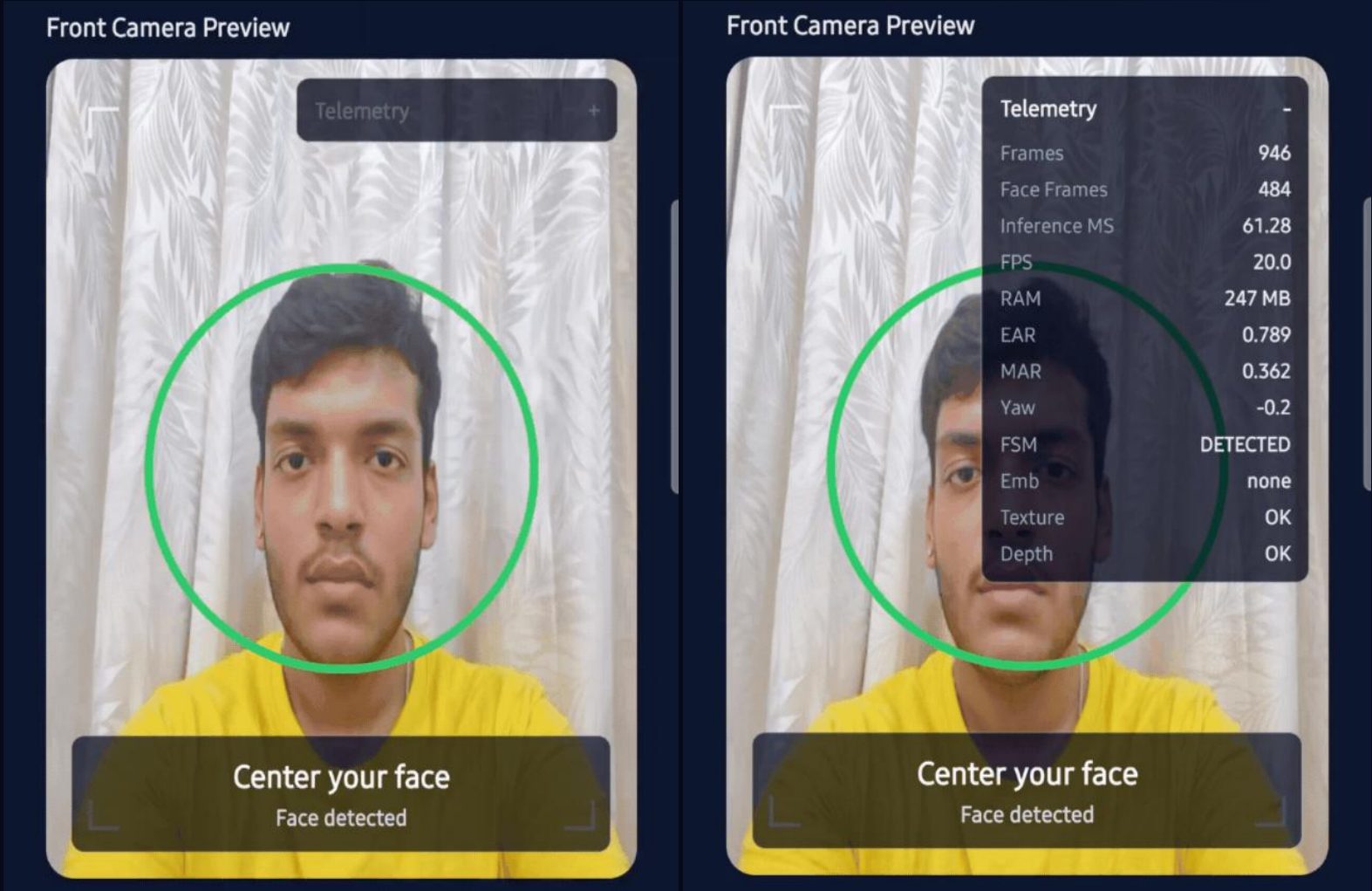


+3.2 %
Over Baseline

Multi-Modal Liveness Detection

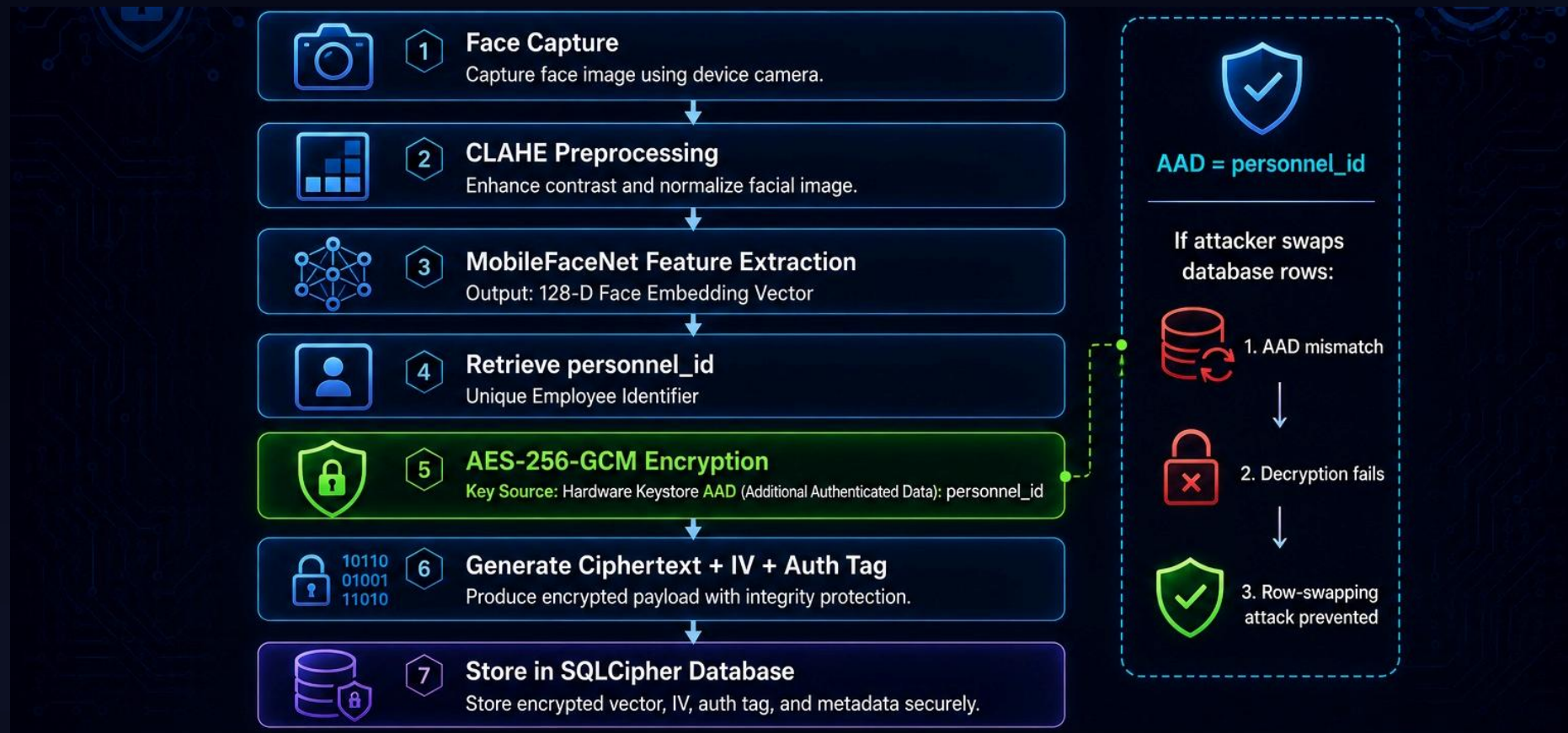
- ✓ 5-State Challenge FSM: Interactive user guide requiring instant action challenge prompts (blink, smile, head tilt) inside a random 4-second safety window.
- ✓ Passive Anti-Spoofing Sensors: Fast Fourier Transform (FFT) high-frequency texture checks scanning for micro-patterns of screens or printed paper.
- ✓ Tactile Synchronization: Built-in haptic confirmation patterns to guide field personnel elegantly through extreme glare and outdoor noise.

The GIF are from the actual app in testing phases showing real time data acquisition



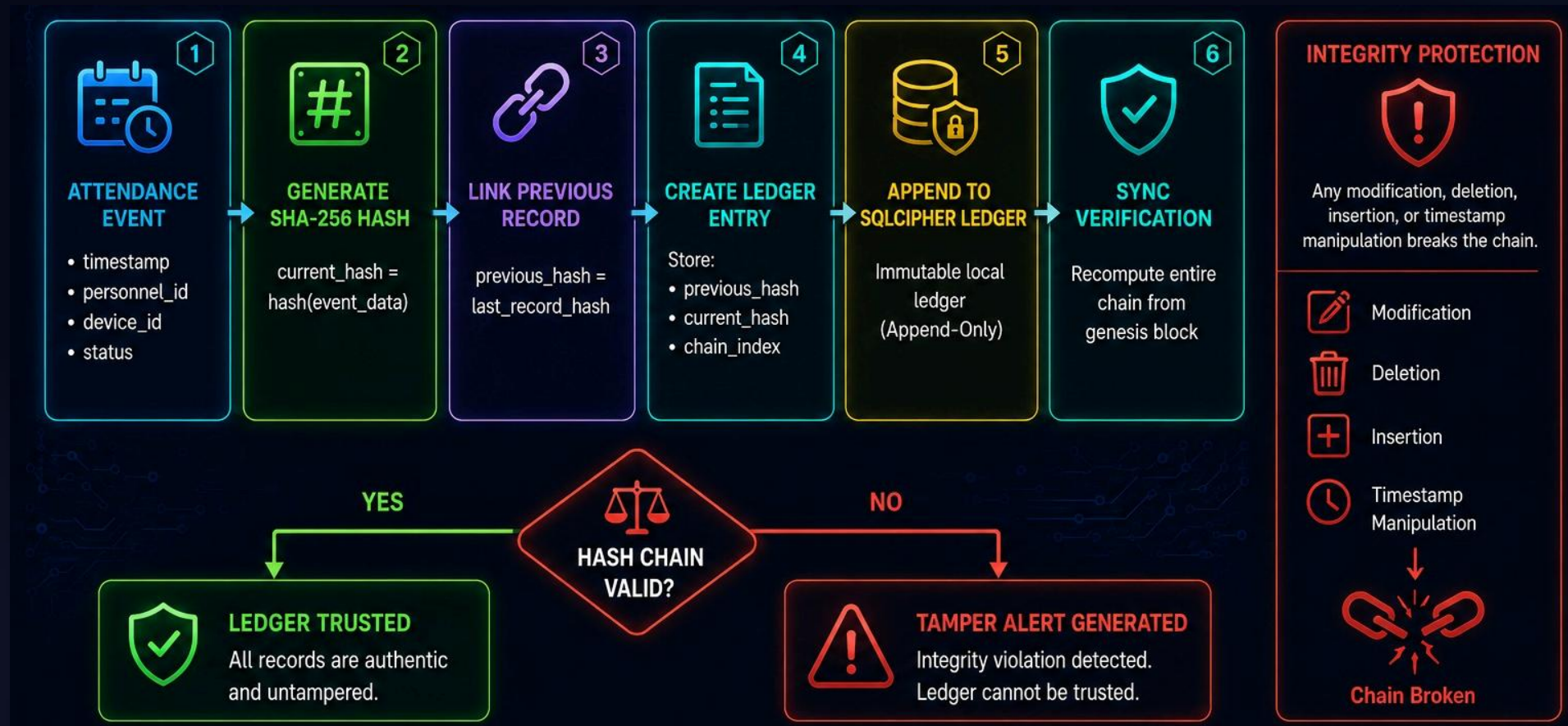
Military-Grade Device Security

- ✓ **Vector-Bound Encryption:** Standard biometric vectors are natively encrypted using AES-256-GCM. Uses individual identifiers as AAD, locking biometric hashes completely.



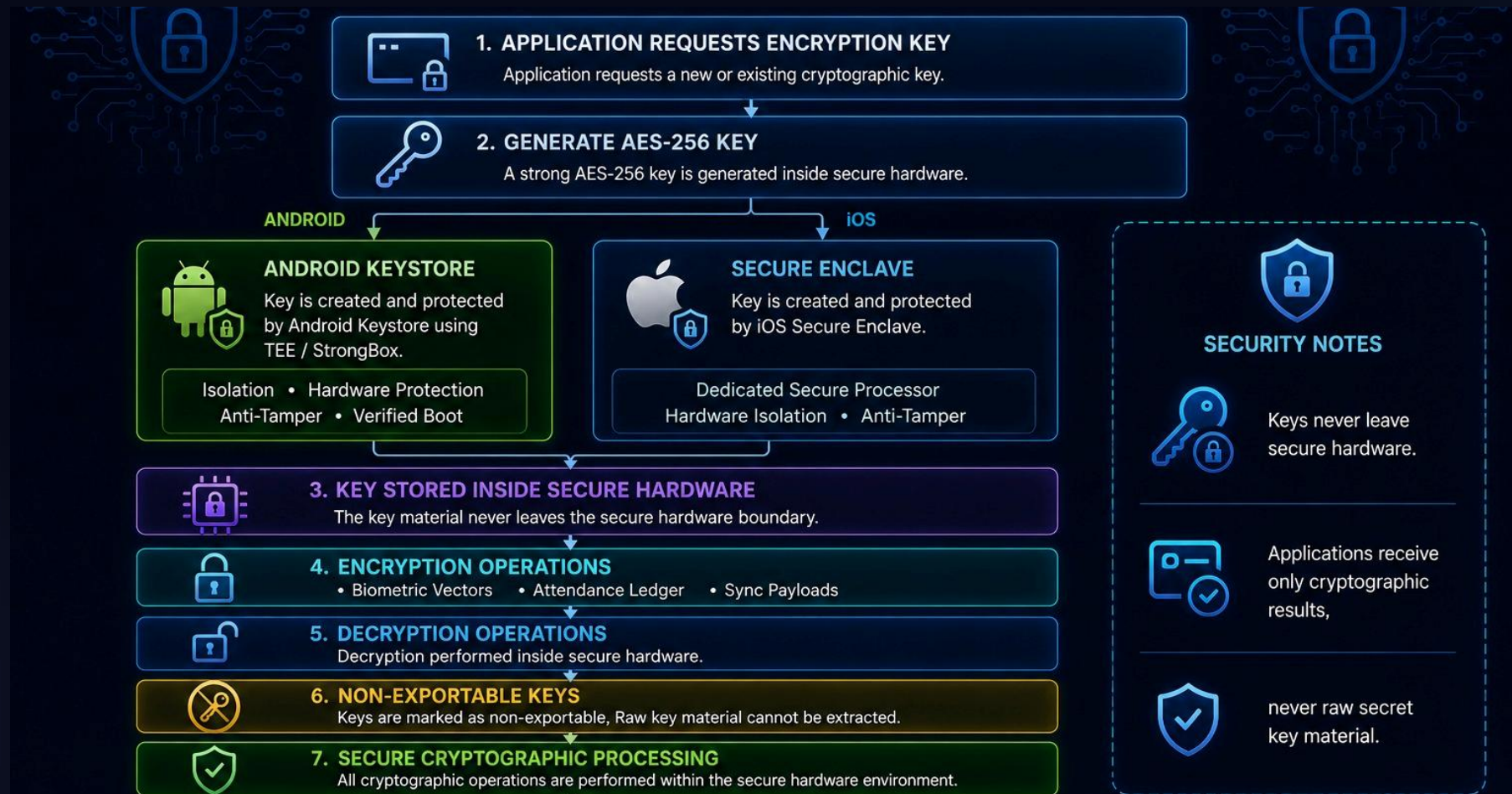
Military-Grade Device Security

- ✓ **Blockchain-Style Ledger:** Every offline attendance event is cryptographically hashed to the preceding event (SHA-256), identifying local data tampering attempts.



Military-Grade Device Security

- ✓ TEE Hardware-Backed Keys: Active keys generated physically inside device Hardware Security Zones (Android StrongBox or iOS SecureEnclave).



Enterprise Scalability & Sync

Operational Vector	Technical Implementation	Core Business Benefit
Local Database Search	Locality-Sensitive Hashing (LSH) 24-bit bucket projection	Limits linear lookup scans to sub-5ms across large local registers.
Network Uplink	NDJSON Batched Uploads via Lambda-generated pre-signed S3 URLs	Bypasses middleware overhead entirely, pushing client records directly.
Storage Management	Automated Atomic Sync & Purge Engine	Safely cleans successfully acknowledged local transactions older than 7 days.

Project Repository & Build Links

Repository: <https://github.com/ShauryaNCode/nayan>

APK (Android) : [Release v1.0.0 · ShauryaNCode/nayan](#)

IOS : `npx react-native run-ios`

(none of our team members use macbooks :))